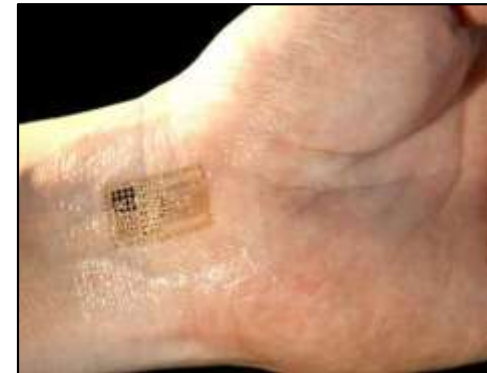




Bio-integrated Systems

More than Moore: Bio-integration



Past

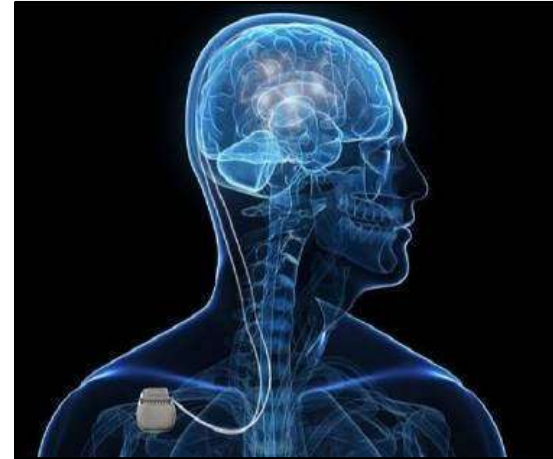
Present

Future

Applications



Biology



Clinical



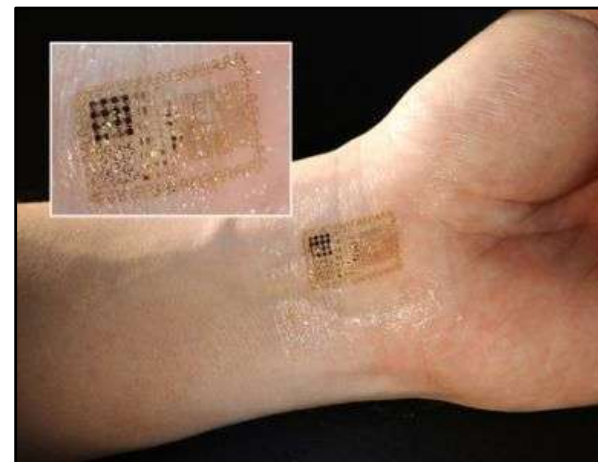
Entertainment

Wearable Devices



temperature, strain
EEG, ECG, EMG
healing, sweat

...

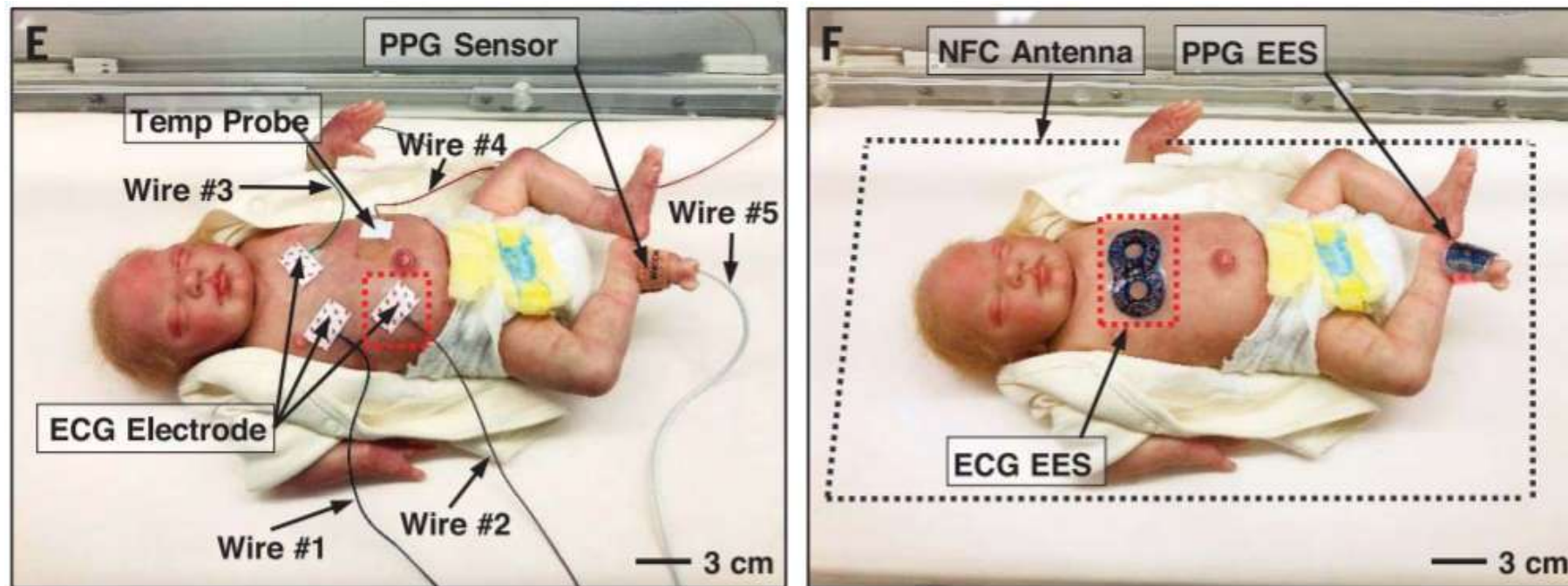


D. Kim, *et al.*, *Science* **333**, 838 (2011)



W. Gao, *et al.*, *Nature* **529**, 509 (2016)

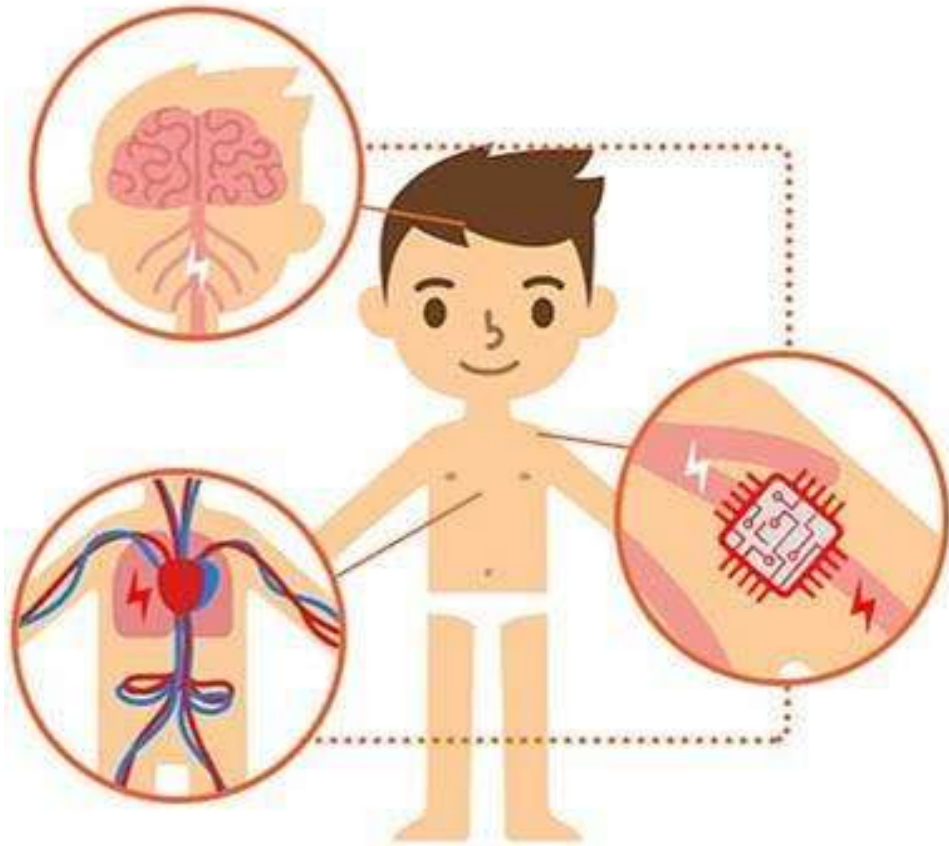
Wearable Devices



neonatal health monitoring

Implantable Devices

Bioelectronic 'Medicine'



Diagnostics & Therapeutics



GSK Australia

[Video](#)

Challenges

■ Conventional electronics and photonics



■ Biological systems



innovations:

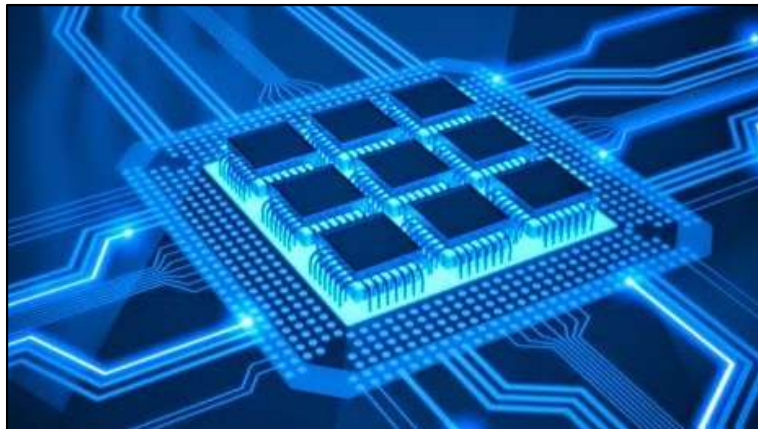
- *materials*

- *devices*

- *systems*

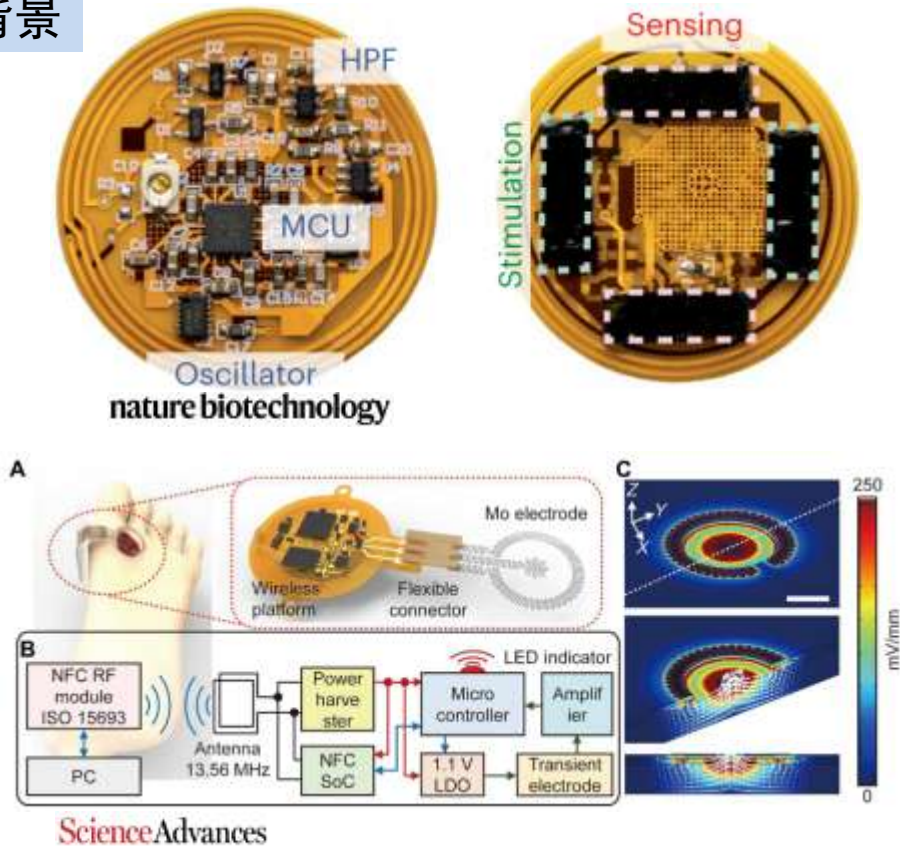
- ...

Interdisciplinary Research



项目概况

背景



应用：柔性可穿戴电子系统，面向生物医疗应用

方法路线

微小型化、高度集成化电子系统

需求应用匹配

生物医疗应用

当前主要研究内容

供能模块：射频、电磁感应无线
传感模块：阻抗分析、电化学分析...
信号传输模块：射频、蓝牙...
电刺激模块：柔性电极...

生物电生理、电化学信号传感
皮肤修复：促进伤口愈合
神经修复：刺激+传感
骨修复：刺激+传感
... ..